SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0501**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

QUESTIONS: 5 **Max Marks50** **Duration :60 Minutes**

Annexure A

Scenario-Based Questions

Code of Conduct:

I **VENNA MEENAKSHI REDDY - 192471048** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Venna Meenakshi Reddy

Annexure A
Scenario-Based Questions

1. **Dental X-Ray Enhancement and Cavity Region Extraction**
A dental clinic uses a digital radiography system to help dentists locate early-stage cavities (caries) in teeth. The X-ray images are dark and low in contrast, with quantum noise and overlapping tooth structures, so small cavity regions blend into the surrounding enamel and dentine.
Question: Apply suitable grey-level transformations (such as negative, log or power-law transformation) and histogram equalization to improve the visibility of the cavity regions. Select an appropriate spatial smoothing filter to suppress noise without blurring the cavity boundaries. Subsequently, apply thresholding or region growing to extract the suspected cavity region, and use edge detection to outline it. Explain the sequence of processing steps and justify the choice of each technique.

2. **Night-Time Vehicle Number Plate Region Extraction**
A traffic police department installs CCTV cameras at a toll gate to capture vehicles at night. The images suffer from very low brightness, headlight glare, sensor noise, and motion blur. The system must locate and extract the number-plate region so that the characters can be read in a later stage.
Question: Apply suitable grey-level transformations and histogram processing to enhance the dark regions while controlling glare. Use appropriate smoothing and sharpening filters (for example, median filtering followed by Laplacian or unsharp masking) to reduce noise and restore character edges. Apply edge detection, line detection and edge linking to locate the rectangular plate boundary, and use thresholding to separate the characters from the plate background. Present the complete processing pipeline and explain how each step contributes to extracting the number-plate region.

3. **Textile Fabric Defect Detection Using Frequency-Domain Processing**
A textile mill inspects woven fabric rolls using a line-scan camera. The fabric has a regular, repeating weave pattern, and defects such as holes, broken threads, oil stains and knots disturb this pattern. The captured images also contain periodic noise from the machine lighting and uneven illumination across the width of the roll.
Question: Apply suitable frequency-domain filtering techniques (low-pass, high-pass, or notch/band-reject filtering) to suppress the periodic noise and the regular weave texture, so that the defects stand out. Use histogram processing to correct the uneven illumination. Subsequently, apply thresholding and boundary detection to segment and outline the defective regions. Explain the order of operations and justify why frequency-domain processing is preferred over spatial-domain processing in this scenario.

4. **Blood Cell Segmentation in Microscopic Images**
A pathology laboratory uses an automated microscope to capture images of stained blood smears for counting red and white blood cells. The images show uneven staining, poor contrast between cells and background, touching or overlapping cells, and noise from dust particles on the slide.
Question: Apply suitable contrast enhancement and histogram processing techniques to improve the separation between cells and background. Use an appropriate spatial filter to remove small noise particles while preserving cell boundaries. Apply global or adaptive thresholding to separate cells from the background, and use region-based segmentation (region growing or region splitting and merging) to separate touching cells. Finally, apply edge detection and boundary detection to outline individual cells. Explain the processing sequence and the role of each technique in obtaining accurate cell regions.

5. **Text Extraction from Degraded Historical Documents**

A state archive is digitising old handwritten manuscripts. The scanned pages contain yellowed paper, ink bleed-through from the reverse side, stains, faded strokes and uneven lighting across the page. The archive requires clean extraction of text regions before handwriting recognition.

Question: Apply suitable grey-level transformations and histogram processing to improve the contrast between faded ink and the background. Select appropriate smoothing and sharpening filters to remove stains and noise while strengthening the pen strokes. Compare global thresholding (such as Otsu's method) with local (adaptive) thresholding for this scenario and apply the more suitable one to separate text from the background. Use edge detection and boundary detection to identify text lines and word regions. Explain the sequence of operations and justify each decision.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

QUESTION 1:

1. Dental X-Ray Enhancement and Cavity Region Extraction

A dental clinic uses a digital radiography system to help dentists locate early-stage cavities (caries) in teeth. The X-ray images are dark and low in contrast, with quantum noise and overlapping tooth structures, so small cavity regions blend into the surrounding enamel and dentine.

Question: Apply suitable grey-level transformations (such as negative, log or power-law transformation) and histogram equalization to improve the visibility of the cavity regions. Select an appropriate spatial smoothing filter to suppress noise without blurring the cavity boundaries. Subsequently, apply thresholding or region growing to extract the suspected cavity region, and use edge detection to outline it. Explain the sequence of processing steps and justify the choice of each technique.

Answer:

1. Scenario Understanding:

A dental clinic wants early-stage caries to be located on digital radiographs. Caries appear as small, slightly darker (radiolucent) patches inside enamel or dentine, but the X-ray is dark, has low contrast, carries signal-dependent quantum noise and shows overlapping tooth structures. The task is therefore a chain of four operations: enhance the contrast, denoise without destroying the lesion, extract the suspect region and outline it for the dentist.

2. Key Issues Identified:

- **Narrow dynamic range:** the histogram is crowded at the dark end (Figure 1.4), so enamel, dentine and cavity differ by only a few grey levels.
- **Quantum (Poisson) noise:** it is amplified by any contrast stretching (Figure 1.3), so denoising must follow enhancement.
- **Small lesion, soft boundary:** a heavy smoothing filter shrinks or erases the cavity, so an edge-preserving filter is needed (Figure 1.7 vs 1.8).
- **Overlapping structures:** absolute grey levels of a cavity overlap with pulp and root regions, so a single global threshold gives false alarms; local contrast must be used.

3. Processing Pipeline and Justification of Each Technique:

Step	Technique	Role in the pipeline and why it is chosen
1	**Power-law (gamma) transform, $s = c \cdot r^{\gamma}$, $\gamma = 0.5$**	The image is dark, so $\gamma < 1$ expands the dark grey levels and lifts enamel/dentine detail. It is preferred to the log transform because γ is tunable and does not over-stretch bright enamel. A negative image ($s = L-1-r$) is kept only as a display aid, since a radiologist can then see lesions as bright spots (Figure 1.6).

Step	Technique	Role in the pipeline and why it is chosen
2	**Histogram equalisation**	The cumulative histogram is used as the mapping so the grey levels spread over 0–255 (Figure 1.4→1.5). This increases the enamel–cavity difference. It also boosts noise, which is why the quantitative steps below use the cleaner power-law output.
3	**Median filter (5×5)**	Quantum noise is impulsive and the cavity boundary is a step edge. The median replaces each pixel by the neighbourhood median, so noise is removed while edges stay sharp. A Gaussian or mean filter averages across the boundary and blurs it (Figure 1.7).
4	**Local-contrast map (background minus image)**	A heavily smoothed copy (Gaussian $\sigma = 10$) is subtracted from the denoised image inside the tooth region. A cavity is darker than its local surroundings, so it becomes a bright peak in Figure 1.9, while smooth enamel/dentine gradients cancel.
5	**Thresholding / region growing**	The map is thresholded at $T = \mu + 3.5\sigma$ of the tooth interior; a closing operation and an area filter remove specks (Figure 1.10). Region growing from the brightest seed with a tolerance of about 5 % is an equivalent alternative that follows the lesion shape.
6	**Edge detection (Canny)**	Canny (Gaussian smoothing, gradient, non-maximum suppression, hysteresis) gives a thin, closed one-pixel contour of the extracted region (Figure 1.11). It is drawn in red over the enhanced image for the dentist (Figure 1.12).

FIGURES:

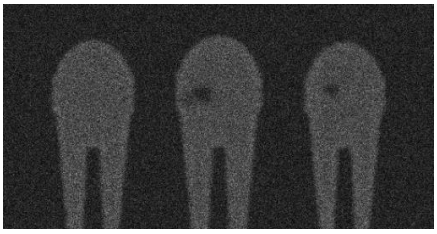


Figure 1.1
Original: dark, low-contrast, noisy

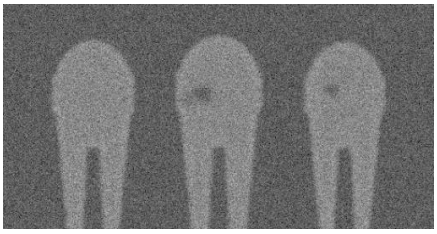


Figure 1.2
Power-law transform (γ=0.5)

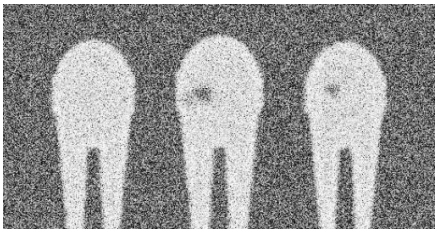


Figure 1.3
Histogram equalisation

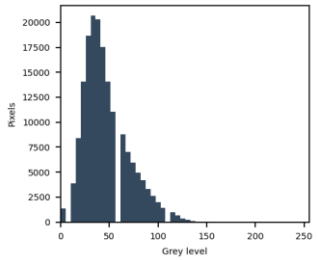


Figure 1.4

Histogram of (a)

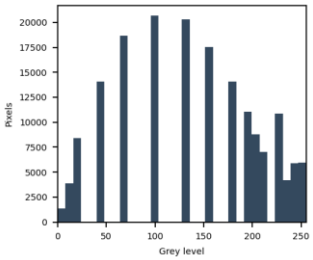


Figure 1.5

Histogram of (c)

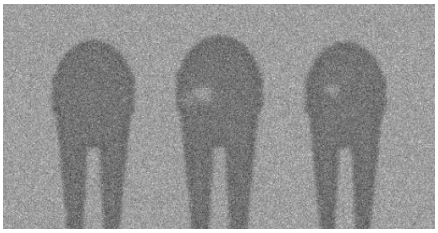


Figure 1.6

Negative of (b)

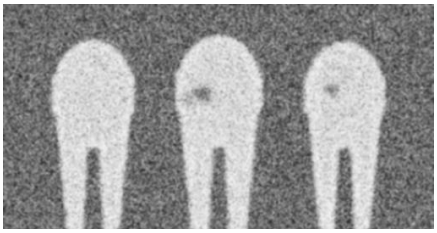


Figure 1.7

Gaussian 5x5: boundaries blur

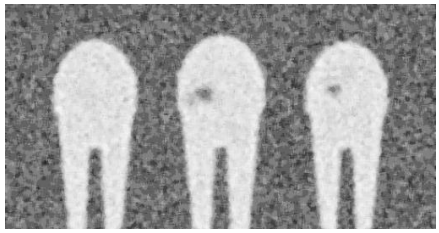


Figure 1.8

Median 5x5: noise removed, edges kept

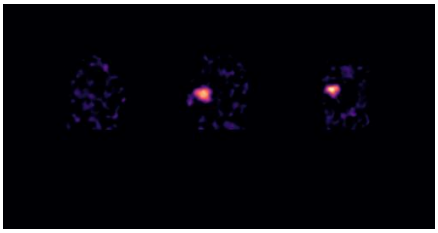


Figure 1.9

Local-contrast (dark-spot) map



Figure 1.10

Thresholded cavity mask

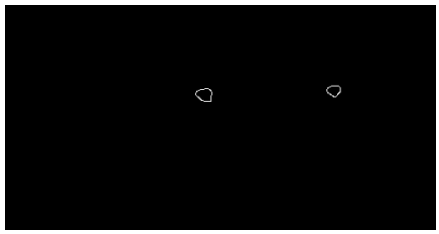


Figure 1.11

Canny edges of cavity region

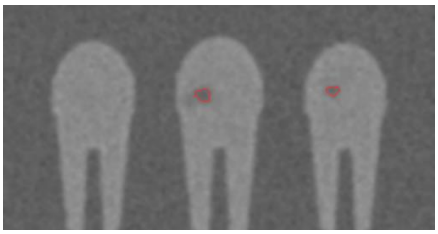


Figure 1.12

Cavity outline on enhanced image

4. Order of operations: Contrast is raised first so that faint structures become measurable; noise is removed next because enhancement amplifies it; segmentation follows on the clean image; edge detection is applied last, on the binary mask, so that a clean closed outline is obtained instead of a noisy edge map.

5. Key decisions:

(i) Median instead of Gaussian filtering, to protect the lesion boundary;

(ii) local-contrast thresholding instead of a global threshold, because healthy dentine and pulp have grey levels similar to the cavity;

(iii) Canny instead of Sobel or Laplacian alone, because of its noise robustness and thin edges.

Result:

Two suspected cavities (middle and right tooth) are extracted and outlined in Figure 1.12, while the healthy left tooth produces no detection. The dentist can therefore inspect only the flagged regions.

QUESTION 2:

2. Night-Time Vehicle Number Plate Region Extraction

A traffic police department installs CCTV cameras at a toll gate to capture vehicles at night. The images suffer from very low brightness, headlight glare, sensor noise, and motion blur. The system must locate and extract the number-plate region so that the characters can be read in a later stage.

Question: Apply suitable grey-level transformations and histogram processing to enhance the dark regions while controlling glare. Use appropriate smoothing and sharpening filters (for example, median filtering followed by Laplacian or unsharp masking) to reduce noise and restore character edges. Apply edge detection, line detection and edge linking to locate the rectangular plate boundary, and use thresholding to separate the characters from the plate background. Present the complete processing pipeline and explain how each step contributes to extracting the number-plate region.

Answer:

1. Scenario Understanding:

A toll-gate CCTV camera captures vehicles at night. The frame is very dark, the headlights saturate (glare), the sensor adds noise and vehicle motion blurs the characters. The system must find the rectangular number plate, cut it out and separate the characters from the plate background so that a later OCR stage can read them.

2. Key Issues Identified:

- **Extreme dynamic range:** the plate is dark while the headlights are saturated; global enhancement makes the glare bloom (Figure 2.3).
- **Sensor noise:** it creates many false edges and is amplified by sharpening, so it must be filtered first.
- **Motion blur:** the horizontal blur weakens character edges, so a sharpening step is required before edge detection.
- **Cluttered edges:** the bumper, grille and windscreen also give straight edges, so the plate must be validated geometrically (aspect ratio, character-edge density).

3. Processing Pipeline and Justification of Each Technique:

Step	Technique	Role in the pipeline and why it is chosen
1	**Power-law transform (γ = 0.45)**	Lifts the very dark plate region and compresses the range, so the plate becomes visible without a fixed gain (Figure 2.2). A log transform would behave similarly.
2	**Histogram processing: CLAHE**	Global equalisation lets the bright glare dominate the histogram and the headlight halos bloom (Figure 2.3). Contrast-limited adaptive equalisation works on 6×6 tiles with a clip limit of 3, so each tile is stretched locally and the clip limit caps both glare and noise amplification (Figure 2.4).

Step	Technique	Role in the pipeline and why it is chosen
3	**Median filter (5×5)**	Removes salt-and-pepper style sensor noise while preserving the plate border. It is applied before sharpening because sharpening would amplify noise.
4	**Unsharp masking (or Laplacian sharpening)**	f_sharp = f + k(f − f_blur) adds back the high-frequency detail removed by blur. It restores the character and plate edges weakened by motion blur (Figure 2.6). Laplacian sharpening, f − 0.6□²f, gives a similar result.
5	**Edge detection (Canny)**	Gradient-based detection with hysteresis gives a thin edge map. The plate border and character strokes are strong, long edges (Figure 2.7).
6	**Line detection (Hough) and edge linking**	The Hough transform detects near-horizontal straight lines ($\|\theta\| < 4°$) such as the plate top and bottom borders (Figure 2.8). Collinear segments with the same y and a gap below 25 px are linked into one line, which bridges the breaks caused by glare and blur (Figure 2.9).
7	**Plate boundary decision**	Two linked lines 35–90 px apart with an aspect ratio of 2.2–6.5 form a plate candidate; the candidate with the highest vertical-edge density (character strokes) is kept and drawn in green (Figure 2.10).
8	**Cropping and Otsu thresholding**	The plate is cropped (Figure 2.11) and Otsu picks the threshold between the two histogram modes (plate and ink). The characters become black on a white background for OCR (Figure 2.12).

FIGURES:



Figure 2.1
Original night frame



Figure 2.2
Power-law (γ=0.45)



Figure 2.3
Global equalisation: glare blooms



Figure 2.4
CLAHE after power-law



Figure 2.5
Median filter 5x5



Figure 2.6
Unsharp masking



Figure 2.7
Canny edge map



Figure 2.8
Hough line detection (H/V lines)



Figure 2.9
Edge linking (merged segments)



Figure 2.10
Plate boundary located

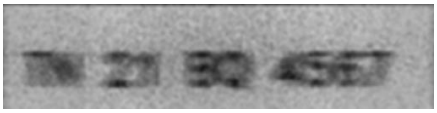


Figure 2.11
Extracted plate region

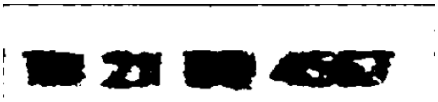


Figure 2.12
Otsu threshold: characters

4. How each step contributes: The grey-level and histogram steps make the plate visible, median filtering removes the noise that would break the edge map, sharpening restores the strokes, line detection and edge linking turn isolated edge pixels into a rectangle, and thresholding finally isolates the characters.

5. Key decisions: CLAHE is chosen over global equalisation to control glare; median filtering is chosen before sharpening so that noise is not amplified; geometric validation (two long parallel lines, aspect ratio, character-edge density) is chosen because edge detection alone also fires on the bumper and grille.

Result:

The plate is located correctly in Figure 2.10, away from the headlight glare, and the characters are separated in Figure 2.12. Because the plate is cropped first, Otsu works on a simple two-class histogram.

QUESTION 3:

3. Textile Fabric Defect Detection Using Frequency-Domain Processing

A textile mill inspects woven fabric rolls using a line-scan camera. The fabric has a regular, repeating weave pattern, and defects such as holes, broken threads, oil stains and knots disturb this pattern. The captured images also contain periodic noise from the machine lighting and uneven illumination across the width of the roll.

Question: Apply suitable frequency-domain filtering techniques (low-pass, high-pass, or notch/band-reject filtering) to suppress the periodic noise and the regular weave texture, so that the defects stand out. Use histogram processing to correct the uneven illumination. Subsequently, apply thresholding and boundary detection to segment and outline the defective regions. Explain the order of operations and justify why frequency-domain processing is preferred over spatial-domain processing in this scenario

Answer:

1. Scenario Understanding:

A line-scan camera inspects woven fabric. The weave is a regular repeating pattern, the machine lighting adds periodic noise (bands) and the illumination changes across the roll width. Defects (hole, oil stain, broken thread, knot) are non-periodic disturbances of this pattern. The aim is to remove everything periodic so that only the defects remain, then to segment and outline them.

2. Key Issues Identified:

- **Regular weave texture:** it fills the whole image in the spatial domain but appears as a few sharp peaks in the Fourier spectrum (Figure 3.2).
- **Periodic lighting noise:** horizontal bands and diagonal banding form extra isolated spectral spikes.
- **Uneven illumination:** a slow, multiplicative brightness change across the width that would defeat any single threshold (Figure 3.5).
- **Different defect types:** a small dark hole, a broad faint stain, a thin broken thread and a bright knot need a detector that is not tied to one size or polarity.

3. Processing Pipeline and Justification of Each Technique:

Step	Technique	Role in the pipeline and why it is chosen
1	**2-D Fourier transform**	The centred log-magnitude spectrum is computed (Figure 3.2). Weave and noise appear as isolated peaks at ±50 cycles/image on both axes and on the diagonal (weave), ±20 along the vertical axis (lighting bands) and (±3, ±5) (diagonal banding). Defects spread their energy over a broad range.
2	**Notch (band-reject) filter**	A Gaussian notch, $H = \Pi[1 - \exp(-D^2/2\sigma^2)]$ with $\sigma = 3$, is placed at each detected peak and its mirror (Figure 3.3). Only those frequencies are removed, so the DC term and the rest of the spectrum, which carry the defects, are kept.

Step	Technique	Role in the pipeline and why it is chosen
3	**Inverse FFT**	The filtered spectrum is transformed back; the weave and bands vanish and only non-periodic defects remain (Figure 3.4).
4	**Illumination correction (homomorphic) and histogram stretch**	Since f = illumination × reflectance, the log image is high-pass filtered by subtracting its Gaussian low-pass (σ = 45, Figure 3.5). A histogram stretch (0.3–99.7 %) then restores contrast (Figure 3.6).
5	**Thresholding**	The deviation from the background is converted to a robust z-score (median and MAD) and thresholded at 6. Opening and closing remove specks and join the broken thread (Figures 3.7–3.8).
6	**Boundary detection**	Contours of the remaining connected regions are traced and drawn on the original fabric (Figure 3.9). The hole, oil stain, broken thread and knot are all outlined.

FIGURES:

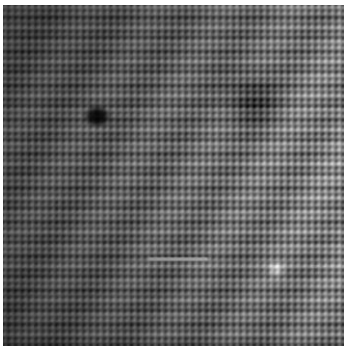


Figure 3.1

Fabric image: weave, stripes, shading

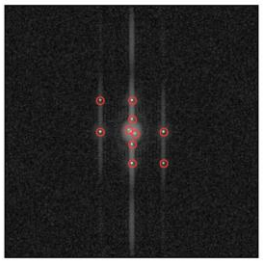


Figure 3.2

Fourier spectrum, peaks marked



Figure 3.3

Notch-reject filter H(u,v)

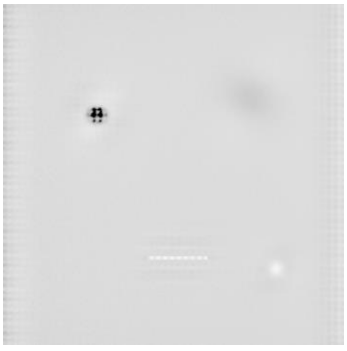


Figure 3.4

After notch filter + homomorphic

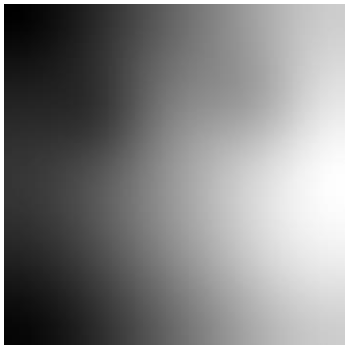


Figure 3.5

Estimated illumination (low-freq)

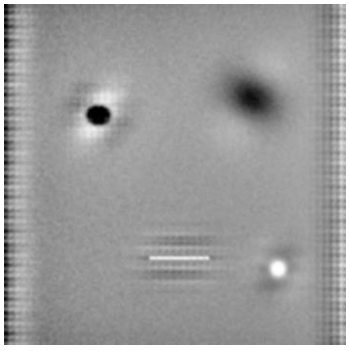


Figure 3.6

Histogram stretch (0.3-99.7%)

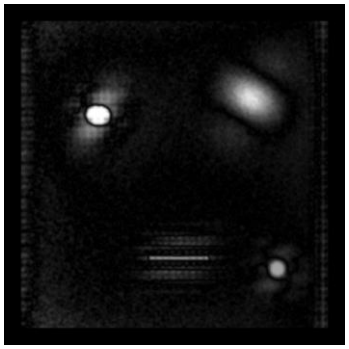


Figure 3.7

Deviation from background



Figure 3.8

Threshold + morphology

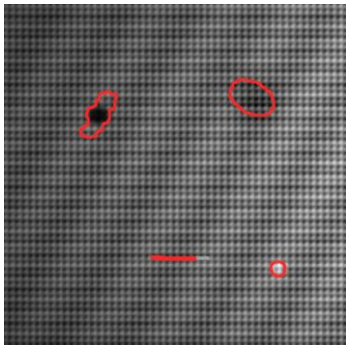


Figure 3.9

Defect boundaries on original

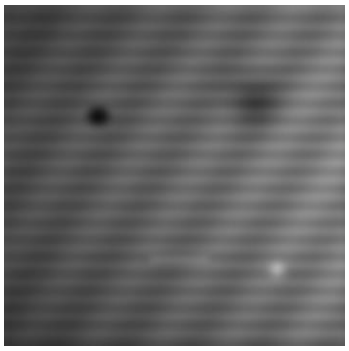


Figure 3.10

Spatial Gaussian only: weave remains out

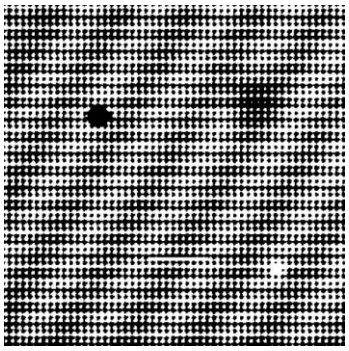


Figure 3.11

Spatial threshold: texture false alarms

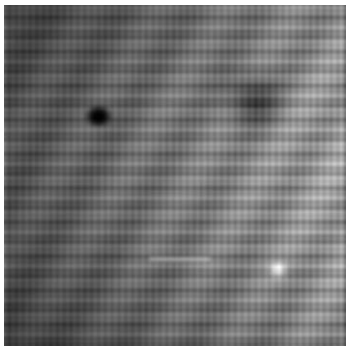


Figure 3.12

Low-pass only: defects blur

4. Order of operations: Notch filtering is done first because the periodic texture and noise must be removed before any thresholding; illumination correction comes next, since the notch filter does not change the slow brightness trend; thresholding and boundary tracing are applied last on a flat, defect-only image.

5. Why the frequency domain: A periodic pattern is compact in frequency (a few points) but spread over every pixel in space, so it can be removed exactly with a few notches without touching the defects. Spatial smoothing (Figure 3.10) leaves the weave, a spatial threshold on the raw texture produces false alarms everywhere (Figure 3.11) and a plain low-pass filter blurs the thin thread and the small knot (Figure 3.12). Spatial filters also cost more for large kernels, whereas the FFT is O(N log N), which suits line-scan speeds.

Result:

All four defects are outlined in Figure 3.9 and the background weave produces no false regions.

QUESTION 4:

4. Blood Cell Segmentation in Microscopic Images

A pathology laboratory uses an automated microscope to capture images of stained blood smears for counting red and white blood cells. The images show uneven staining, poor contrast between cells and background, touching or overlapping cells, and noise from dust particles on the slide.

Question: Apply suitable contrast enhancement and histogram processing techniques to improve the separation between cells and background. Use an appropriate spatial filter to remove small noise particles while preserving cell boundaries. Apply global or adaptive thresholding to separate cells from the background, and use region-based segmentation (region growing or region splitting and merging) to separate touching cells. Finally, apply edge detection and boundary detection to outline individual cells. Explain the processing sequence and the role of each technique in obtaining accurate cell regions.

Answer:

1. Scenario Understanding:

A pathology laboratory images stained blood smears to count red and white blood cells. The images have uneven staining, weak contrast between cells and background, dust particles and cells that touch or overlap. A correct count therefore needs enhancement, noise removal, thresholding and a method that separates touching cells into individual regions.

2. Key Issues Identified:

- **Uneven staining:** the background brightness drifts across the slide, so a single threshold would label part of the background as cells.
- **Poor contrast:** cell and background grey levels are close in a plain grey-scale image.
- **Dust particles:** small dark specks appear as false cells unless removed.
- **Touching or overlapping cells:** they merge into one blob after thresholding (31 blobs in Figure 4.12) and lead to under-counting.

3. Processing Pipeline and Justification of Each Technique:

Step	Technique	Role in the pipeline and why it is chosen
1	**Channel selection and negative**	The stain absorbs green light, so the green channel has the best cell–background contrast (Figure 4.2). Its negative makes cells bright.
2	**Flat-field correction and contrast enhancement**	A large median filter (101×101) estimates the slowly varying background; subtracting it removes the staining gradient. Contrast stretching and CLAHE (4×4 tiles) then separate the cells from the background (Figure 4.3).
3	**Median filter (5×5)**	Dust specks are only 1–3 px wide, so the median removes them. It keeps the cell boundaries, whereas a Gaussian filter would smear the thin gap between touching cells (Figure 4.4).

Step	Technique	Role in the pipeline and why it is chosen
4	**Global (Otsu) thresholding**	After flat-field correction the histogram is clearly bimodal (Figure 4.5), so Otsu's global threshold (T = 116) is sufficient (Figure 4.6). Adaptive thresholding is the fallback if the staining gradient is stronger.
5	**Morphology and hole filling**	Filling holes closes the pale central region of red cells, and an elliptical opening (7×7) removes residual specks (Figure 4.7).
6	**Region-based splitting: distance transform + marker-controlled watershed**	The distance transform (Figure 4.8) has one peak per cell; the peaks become markers (Figure 4.9). Regions grow from the markers until they meet at the necks between cells, which splits touching cells (Figure 4.10): 31 merged blobs become 39 cells.
7	**Edge / boundary detection**	The boundaries of the watershed labels are extracted and drawn on the original smear (Figure 4.11). Each cell now has its own closed outline for counting and for area measurement (small RBCs and large WBCs).

FIGURES:

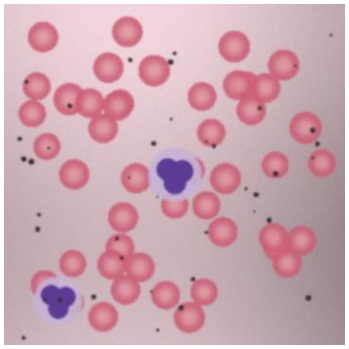


Figure 4.1
Stained blood smear (RGB)

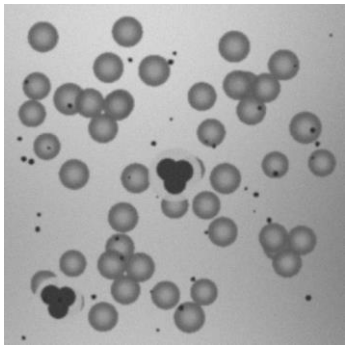


Figure 4.2
Green channel (best contrast)

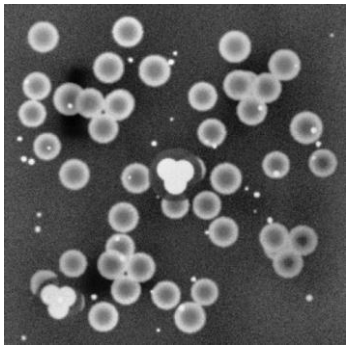


Figure 4.3
Negative, flat-field + CLAHE

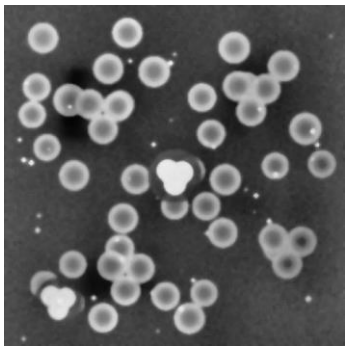


Figure 4.4
Median 5x5: dust removed

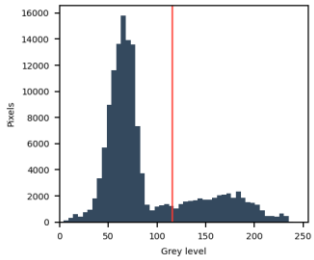


Figure 4.5
Histogram, Otsu T = 116

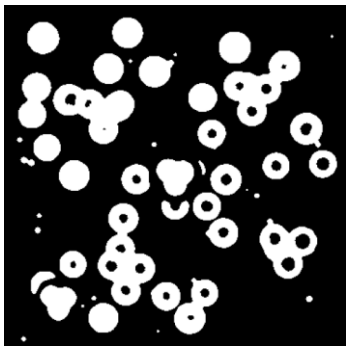


Figure 4.6
Global Otsu threshold

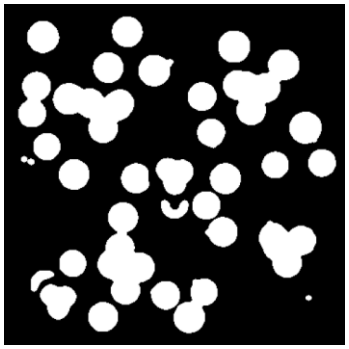


Figure 4.7

Mask after opening + hole filling

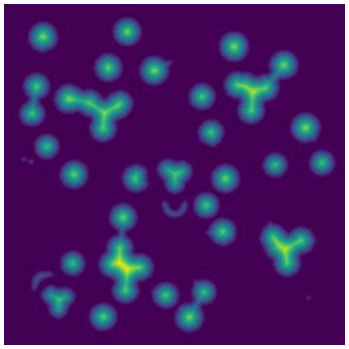


Figure 4.8

Distance transform

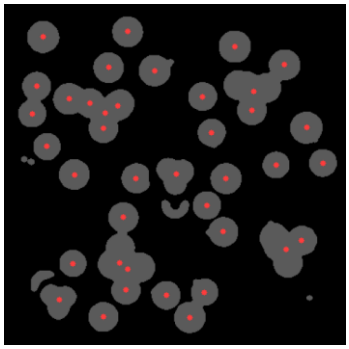


Figure 4.9

Markers = distance peaks



Figure 4.10

Watershed: 39 regions

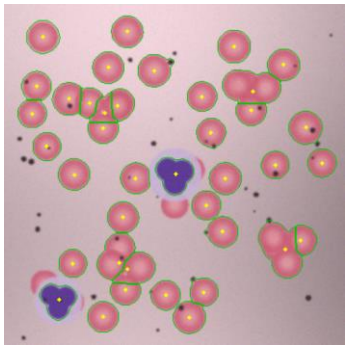


Figure 4.11

Cell boundaries on original



Figure 4.12

Without splitting: 31 blobs

4. Order of operations: Background correction and contrast enhancement come first so that thresholding is reliable; noise is removed before thresholding so that specks do not become objects; splitting touching cells is done on the binary mask, and boundary detection is done last on the separated labels.

5. Key decisions: The green channel is used because it carries the most contrast; the median filter is used because it removes dust without blurring boundaries; the watershed is used because thresholding alone cannot separate cells that touch (Figure 4.12 shows the blobs left unsplit).

Result:

Every cell in Figure 4.11 has its own outline, which gives a more accurate count than the 31 merged blobs. Strongly overlapped cells may still need manual review.

QUESTION 5:

5. Text Extraction from Degraded Historical Documents

A state archive is digitising old handwritten manuscripts. The scanned pages contain yellowed paper, ink bleed-through from the reverse side, stains, faded strokes and uneven lighting across the page. The archive requires clean extraction of text regions before handwriting recognition.

Question: Apply suitable grey-level transformations and histogram processing to improve the contrast between faded ink and the background. Select appropriate smoothing and sharpening filters to remove stains and noise while strengthening the pen strokes. Compare global thresholding (such as Otsu's method) with local (adaptive) thresholding for this scenario and apply the more suitable one to separate text from the background. Use edge detection and boundary detection to identify text lines and word regions. Explain the sequence of operations and justify each decision.

Answer:

1. Scenario Understanding:

A state archive scans old handwritten manuscripts for handwriting recognition. The pages have yellowed paper, bleed-through from the reverse side, stains, faded strokes and uneven lighting. Before recognition, the text regions must be cleaned and separated from the background, and text lines and words must be located.

2. Key Issues Identified:

- **Low contrast:** yellowed paper and faded ink bunch the histogram into a narrow band (Figure 5.3).
- **Stains and bleed-through:** dark blobs and faint mirrored strokes look like ink to a simple threshold.
- **Uneven lighting:** the page brightness changes from left to right, so one global threshold cannot suit the whole page.
- **Thin pen strokes:** they must be strengthened without enhancing the paper grain.

3. Processing Pipeline and Justification of Each Technique:

Step	Technique	Role in the pipeline and why it is chosen
1	**Grey conversion and power-law transform ($\gamma = 1.5$)**	The RGB scan is converted to grey. $\gamma > 1$ darkens the mid-tones, which lowers the paper level and separates the dark ink from the bright background; the result is stretched to the full range.
2	**Histogram processing: CLAHE**	Local adaptive equalisation (6×6 tiles, clip limit 2.5) raises the contrast of faded strokes in each region and compensates part of the lighting drift (Figure 5.4).
3	**Median filter (3×3)**	Removes paper-grain speckle and isolated noise while preserving the thin strokes. Large stains are not removed here; they are handled by local thresholding (Figure 5.5).

Step	Technique	Role in the pipeline and why it is chosen
4	**Unsharp masking**	Subtracting a blurred copy strengthens the pen strokes and the stroke-to-paper gradient (Figure 5.6).
5	**Thresholding: global vs local (see comparison below)**	Local Sauvola thresholding, $T(x,y) = m(x,y)[1 + k(s(x,y)/R - 1)]$ on a 35×35 window, is applied (Figure 5.8). Morphological opening and a small-component filter then clean the mask (Figure 5.9).
6	**Edge detection**	Canny edges of the cleaned mask give the outline of every stroke (Figure 5.10). They are used to check that strokes remain connected.
7	**Boundary detection of lines and words**	The row profile of text pixels shows one peak per text line (Figure 5.11). Horizontal dilation joins the strokes of a line (70×3) or of a word (13×5), and contours give the bounding boxes: blue for lines, green for word regions (Figure 5.12).

FIGURES:

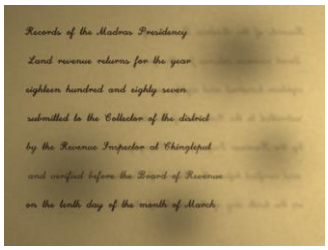


Figure 5.1
Scanned manuscript (RGB)

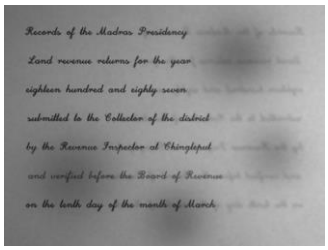


Figure 5.2
Grey-scale: low contrast

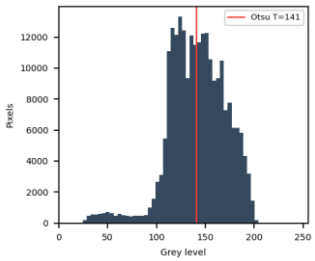


Figure 5.3
Histogram of (b)

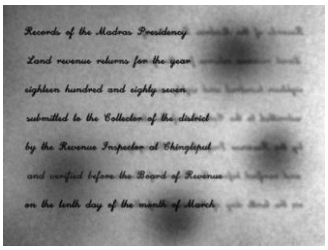


Figure 5.4
Power-law (γ=1.5) + CLAHE

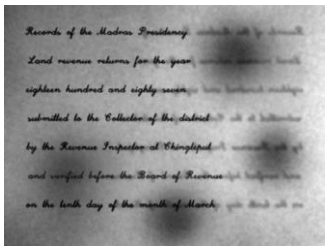


Figure 5.5
Median 3x3 (stains, speckle)

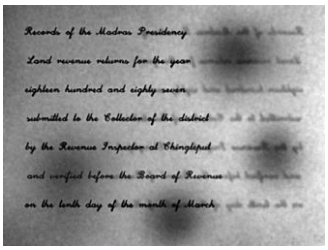


Figure 5.6
Unsharp masking (strokes)

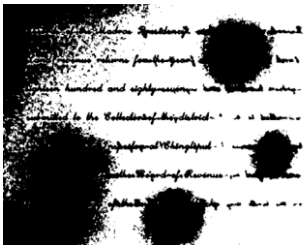


Figure 5.7

Global Otsu: stains and shading fail

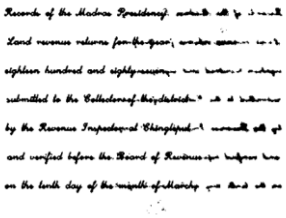


Figure 5.8

Local (Sauvola) threshold: clean

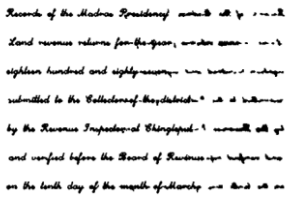


Figure 5.9

After morphological clean-up

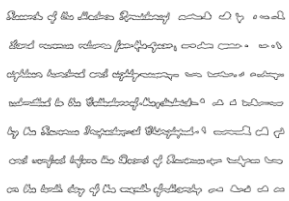


Figure 5.10

Canny edges of strokes

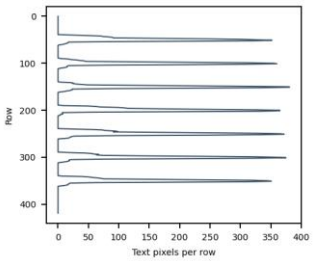


Figure 5.11

Row profile: text-line peaks

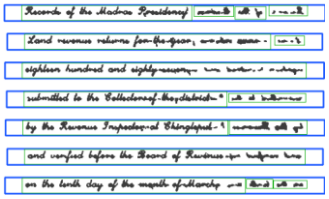


Figure 5.12

7 lines / 15 word regions

Table 5.1: Comparison of Global (Otsu) and Local (Sauvola) Thresholding

Criterion	Global thresholding (Otsu)	Local / adaptive thresholding (Sauvola)
Threshold	One value for the whole page (T = 141)	A separate value for every pixel from its window mean and deviation
Uneven light, stains	Fails: the dark left edge and the stains turn black (Figure 5.7)	Follows the local background, so stains and shading are ignored (Figure 5.8)
Faded strokes	Lost where the ink is lighter than T	Kept, because they are compared with their own neighbourhood
Cost and use	Very fast, suitable for evenly lit scans	Slightly slower, suitable for degraded pages

4. Order of operations: Contrast is enhanced first, noise is filtered second, strokes are sharpened third, and only then is the page binarised; sharpening before filtering would amplify the paper grain, and thresholding before enhancement would lose faded strokes.

5. Key decision: Local thresholding is chosen because the histogram of the page is single-peaked (Figure 5.3) and the lighting is uneven, so Otsu's one threshold cannot separate ink from the stains and the dark edge, while Sauvola uses the local mean and deviation and keeps the text only.

Result:

The text is cleanly separated (Figure 5.9) and 7 text lines are located by the row profile and the bounding boxes (Figure 5.12), ready for handwriting recognition.